

Classifying Questions by Difficulty

A 2003 study by Professors Janet Metcalfe (Columbia University) and Nate Kornell (Williams College) analyzed how students allocated time to study based given a certain study time. It suggests that students attempt to study the easiest questions first, and then spending the most time on medium-level questions, when preparing for exams. As students generally grow more familiar with the content, they grow more confident in their ability to solve difficult problems. At Boilerexams, exam questions are published without indication of their difficulty, as professors do not have a standardized method of indicating how difficult a question on an exam may be. We looked to calculate a metric of difficulty for each question available within our database of CS 25100 exams. In doing so, we enable students with the ability to study course content with progressive difficulty.

At Boilerexams, we collect data on how accurately students answer questions, if a student watched the video solution to the question, and how long they watched the videos for. We used exam date data from University of Purdue University's database of courses and exams, to compare when questions are answered to the nearest exam date for a course.

In the development of our analysis, we calculate the percentage error rate of each question, average time spent on the question, and the percentage of users who watched the video. We then correlate each submission with how far away the closest exam was. From this data, we used an array of calculations to determine exam difficulty.

$$\text{Error Rate} = \frac{\text{Total Incorrect Submissions}}{\text{Total Submissions}}$$

$$\text{Average Time Spent} = \frac{\text{Cumulative Time Spent (CTS) on Question from All Users}}{\text{Total Number of Submissions}}$$

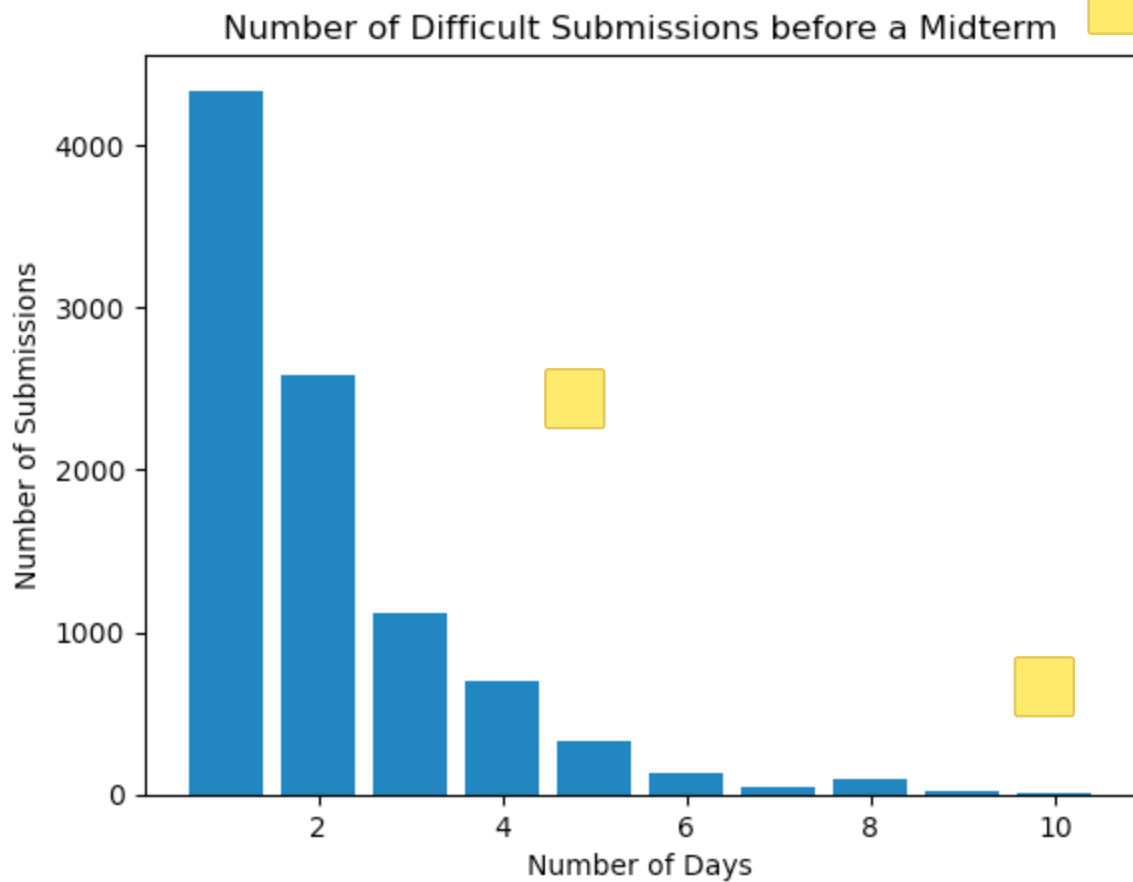
$$\text{Video Watch Rate} = \frac{\text{CTS on Watching Video Solution}}{\text{CTS on Question} + \text{CTS on Watching Video Solution}}$$

We then set weights to each of these metrics. Our final difficulty value for each question is calculated by the following:

Total Difficulty = $(0.6 \times \text{Normalized Error Rate}) + (0.3 \times \text{Average Time Spent}) + (0.1 \times \text{Video Watch Rate})$
The question is then categorized by the percentile of its difficulty score as follows:

Difficulty	Range
Easy	[0, 30)
Medium	[30, 70]
Hard	(70, 100]

We finally plot the frequency of submissions based on their difficulty.



We find that as an exam approaches, the number of 'difficult' questions answered drastically increases. This conclusion is consistent with our hypothesis that as the exam date approaches, students generally practice more difficult questions.

Our next steps are to normalize our data for a variety of different courses and determine performance in different subject areas entirely. We can also determine which topics are regarded as more difficult and explore if users improved accuracy on similar questions with the same topic after watching a Boilerexams video on any given exam question.